$\langle \phi | \hat{H} | \phi \rangle = 2(\alpha + \beta - 2) - g$

$\langle \phi | \hat{H} | \phi_i^a \rangle = -\frac{1}{2} g$

$\langle \phi | \hat{H} | \phi_{ij}^{ab} \rangle = 0$

	$\|12\rangle$	$\|13\rangle$	$\|14\rangle$	$\|23\rangle$	$\|24\rangle$	$\|34\rangle$
$\|12\rangle$	$2-g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	0
$\|13\rangle$	$-\frac{1}{2}g$	$4-g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	0	$-\frac{1}{2}g$
$\|14\rangle$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$6-g$	0	$-\frac{1}{2}g$	$-\frac{1}{2}g$
$\|23\rangle$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	0	$6-g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$
$\|24\rangle$	$-\frac{1}{2}g$	0	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$8-g$	$-\frac{1}{2}g$
$\|34\rangle$	0	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$10-g$

3.

	$\|12\rangle$	$\|13\rangle$	$\|14\rangle$	$\|23\rangle$	$\|24\rangle$
$H =$ $\|12\rangle$	$2-g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$
$\|13\rangle$	$-\frac{1}{2}g$	$4-g$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	0
$\|14\rangle$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$6-g$	0	$-\frac{1}{2}g$
$\|23\rangle$	$-\frac{1}{2}g$	$-\frac{1}{2}g$	0	$6-g$	$-\frac{1}{2}g$
$\|24\rangle$	$-\frac{1}{2}g$	0	$-\frac{1}{2}g$	$-\frac{1}{2}g$	$8-g$

Hartree-Fock

$$\hat{H} = \hat{H}_0 - \hat{V}$$

$$\hat{H}_0 = \sum_{p\sigma} (p-1)\, a^+_{p\sigma} a_{p\sigma}$$

$$a^+_{p\sigma} a_{p\sigma} = 1 \quad (p < F)$$

$$a^+_{p\sigma} a_{p\sigma} = 0 \quad (p > F)$$

We must reorder anihalation/creation opperators so they are normal ordered with respect to the ground state $|12\rangle$ instead of vacuum $|0\rangle$.

since $|12\rangle$ are both states under fermi level occupied, we want all anihalation opperators to the left:

$$a_{p\sigma} a^+_{p\sigma} = 0 \quad (p < F)$$

$$\hat{H}_0 = \sum_{p\sigma} (p-1)(\delta_{pF} - a_{p\sigma} a^+_{p\sigma})$$

$$= \sum_{\substack{p<F \\ \sigma}} (p-1) - \sum_{p\sigma} (p-1)\, a_{p\sigma} a^+_{p\sigma}$$

$p < F = i \quad , \quad p > F = a$

$$\hat{H}_0 = \sum_{i\sigma} (i-1) - \sum_{p\sigma} (p-1) \alpha_{p\sigma} \alpha^{+}_{p\sigma}$$

$$= \sum_{i} 2(i-1) - \sum_{p\sigma} (p-1) \alpha_{p\sigma} \alpha^{+}_{p\sigma}$$

$$= 2(1-1) + 2(2-1) - \sum_{p\sigma} (p-1) \alpha_{p\sigma} \alpha^{+}_{p\sigma}$$

$$= 2 - \sum_{p\sigma} (p-1) \alpha_{p\sigma} \alpha^{+}_{p\sigma}$$

$$= 2 + \sum_{p\sigma} (p-1) N(\alpha^{+}_{p\sigma} \alpha_{p\sigma})$$

$$\hat{V} = -\frac{1}{2} g \sum_{pq} \alpha^{+}_{p+} \alpha^{+}_{p-} \alpha_{q-} \alpha_{q+}$$

$$\alpha^{+}_{p+} \alpha^{+}_{p-} \alpha_{q-} \alpha_{q+} = N(\alpha^{+}_{p+} \alpha^{+}_{p-} \alpha_{q-} \alpha_{q+})$$

Using Wicks theorem

$$+ N(\alpha^{+}_{p+} \underbrace{\alpha^{+}_{p-} \alpha_{q-}} \alpha_{q+})$$

$$+ N(\alpha^{+}_{p+} \underbrace{\alpha^{+}_{p-} \alpha_{q+}} \alpha_{q-})$$

$$+ N(\underbrace{\alpha^{+}_{p+} \alpha^{+}_{p-}} \alpha_{q-} \alpha_{q+})$$

$$+ N(\underbrace{\alpha^{+}_{p+} \alpha_{q-}} \alpha^{+}_{p-} \alpha_{q+})$$

$$+ N(\underbrace{\alpha^{+}_{p+} \alpha_{q+}} \alpha^{+}_{p-} \alpha_{q-})$$

$$+ N(\underbrace{\alpha^{+}_{p-} \alpha_{q-}} \alpha^{+}_{p+} \alpha_{q+})$$

$$+ N(\underbrace{\alpha^{+}_{p-} \alpha_{q+}} \alpha^{+}_{p+} \alpha_{q-})$$

$$+ N(\alpha^{+}_{p+} \alpha^{+}_{p-} \alpha_{q-} \alpha_{q+})$$

$$- N(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+})$$

$$+ \delta_{pq}\delta_{pq<F}\, N(a^\dagger_{p+} a_{q+})$$

$$+ 0$$

$$+ 0$$

$$+ \delta_{pq}\delta_{pq<F}\, N(a^\dagger_{p-} a_{q-})$$

$$+ \delta_{pq}\delta_{pq<F}$$

$$+ 0$$

$$\hat{V} = -\frac{1}{2}g\left[\sum_i 1 + N(a^\dagger_{i+} a_{i+}) + N(a^\dagger_{i-} a_{i-}) + \sum_{pq} N(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+})\right]$$

$$\hat{V} = -g + \sum_{i\sigma} N(a^\dagger_{i\sigma} a_{i\sigma}) + \sum_{pq} N(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+})$$

$$\hat{H} = (2-g) + \sum_{p\sigma}(p-1) N(a^\dagger_{p\sigma} a_{p\sigma})$$

$$-\frac{1}{2}g\sum_{i\sigma} N(a^\dagger_{i\sigma} a_{i\sigma}) - \frac{1}{2}g\sum_{pq} N(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+})$$

$$\hat{H} = E_{ref} + \hat{F}_N + \hat{V}_N$$

$$E_{ref} = 2\text{-}g$$

$$\hat{F}_N = \sum_{p\sigma} (p-1) N(a^\dagger_{p\sigma} a_{p\sigma}) - \frac{1}{2} g \sum_{i\sigma} N(a^\dagger_{i\sigma} a_{i\sigma})$$

$$\hat{V}_N = -\frac{1}{2} g \sum_{pq} N(a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+})$$

Deriving HF equation:

$$|p\rangle = \sum_{\alpha} C_{p\alpha} |\alpha\rangle$$

$$E = \sum_{p} \langle p|h|p\rangle + \frac{1}{2} \sum_{pq} \langle pq|V|pq\rangle_{AS}$$

Inserting original basis:

$$E = \sum_{p} \sum_{\alpha\gamma} C^*_{p\alpha} C_{p\gamma} \langle \alpha|h|\gamma\rangle$$

$$+ \frac{1}{2} \sum_{pq} \sum_{\substack{\alpha\beta \\ \gamma\delta}} C^*_{p\alpha} C^*_{p\beta} C_{p\gamma} C_{ps} \langle \alpha\beta|V|\gamma\delta\rangle$$

inserting constraint:

$$\sum_{\alpha} C_{p\alpha} C^*_{p\alpha} = 1$$

$$\mathcal{L} = E - \sum_{p} \epsilon^{HF}_{p} \left(\sum_{\alpha} C^*_{p\alpha} C_{p\alpha} - 1 \right)$$

Minimizing with respect to $C^*_{p\alpha}$:

$$\frac{\partial \mathcal{L}}{\partial C^*_{p\alpha}} = 0$$

$$\sum_\gamma C_{p\gamma} \langle \alpha | h | \gamma \rangle + \sum_q \sum_{\beta\gamma\delta} C^*_{q\beta} C_{p\gamma} C_{q\delta} \langle \alpha\beta | v | \gamma\delta \rangle_{As} - \epsilon_p^{HF} C_{p\alpha} = 0$$

$$\sum_\gamma \left[\langle \alpha | h | \gamma \rangle + \sum_q \sum_{\beta\delta} C^*_{q\beta} C_{q\delta} \langle \alpha\beta | v | \gamma\delta \rangle_{As} \right] C_{p\gamma} = \epsilon_p^{HF} C_{p\alpha}$$

$$h^{HF}_{\alpha\gamma} = \left[\langle \alpha | h | \gamma \rangle + \sum_q \sum_{\beta\delta} C^*_{q\beta} C_{q\delta} \langle \alpha\beta | v | \gamma\delta \rangle_{As} \right]$$

$$\sum_\gamma h^{HF}_{\alpha\gamma} C_{p\gamma} = \epsilon_p^{HF} C_{p\alpha}$$

$\langle \alpha | h | \gamma \rangle$ → one body part: describes how each particle behaves on its own.

$\langle \alpha\beta | v | \gamma\delta \rangle$ → Two body (direct): describes potential of a particle created by the other particles.

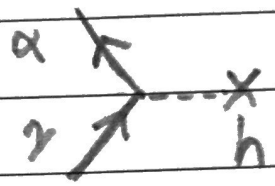
$-\langle \alpha\beta | V | \delta\gamma \rangle$ → Two body (exchange): Consequence of anti-symmetry of fermions.

HF operator in second quantization:

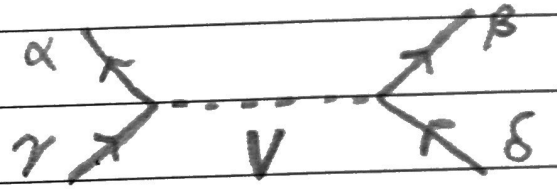
$$\hat{F} = \sum_{\alpha\gamma}^{HP} h_{\alpha\gamma} a_\alpha^\dagger a_\gamma$$

Diagramatic representation:

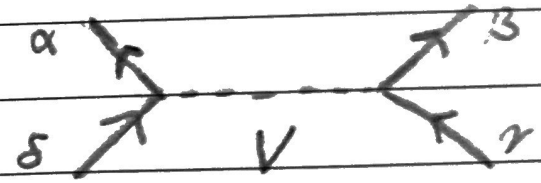
$\langle \alpha | h | \gamma \rangle$ →



$\langle \alpha\beta | V | \gamma\delta \rangle =$



$-\langle \alpha\beta | V | \delta\gamma \rangle =$



Simplifying equation:

$$\langle \alpha | \hat{H}_0 | \beta \rangle = \varepsilon \sum_{p\sigma} (p-1) \langle 0 | a_\alpha a^\dagger_{p\sigma} a_{p\sigma} a^\dagger_\beta | 0 \rangle$$

$$= \varepsilon (p_\alpha - 1) \delta_{\alpha\beta}$$

We can then turn the condition into

$$h^{HF} C = \varepsilon^{HF} C$$

instead of

$$\sum_\gamma h^{HF}_{\alpha\gamma} C_{p\gamma} = \epsilon^{HF}_p C_{p\alpha}$$

since h^{HF} also depends on C, we start with the guess $C^{(0)} = I$. We use this to find h^{HF}, then find $C^{(n)}$ by solving the simlified equation, and repeat until ε^{HF} no longer changes.

Lets do one iteration by hand:

$$C^{(0)} = I$$

$h^{HF}_{\alpha\gamma}$ one body part:

$\langle \alpha | \hat{H}_0 | \gamma \rangle = (R_k - 1)\delta_{\alpha\gamma}$

h^{HF} one body part $= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$

$h^{HF}_{\alpha\gamma}$ two body part:

$$\sum_{P}\sum_{\beta\delta} C^{*}_{P\beta} C_{P\delta} \langle \alpha\beta | V | \gamma\delta \rangle_{AS} =$$

$$\sum_{P}\sum_{\beta\delta} \delta'_{P\beta} \delta'_{P\delta} \langle \alpha\beta | V | \gamma\delta \rangle_{AS} =$$

$$\sum_{P} \langle \alpha P | V | \gamma P \rangle_{AS}$$

replace P with i since $P < F$:

$$\sum_{i} \langle \alpha i | V | \gamma i \rangle_{AS}$$

$\alpha = \gamma$ since V cannot take a state $|\alpha i\rangle$ to $|\gamma i\rangle$ unless $\alpha = \gamma$, so all off diagonal elements become 0.

α and γ must also form a pair with i, so α and γ must be occupied states.

$$h_{ii}^{HF} = \sum \langle \alpha_i | \hat{V}_N | \alpha_i \rangle_{AS}$$

$$= -\frac{1}{2} y \sum_i \sum_{pq} \langle \alpha_i | a_{p+} a_{p-} a_{q-} a_{q+} | \alpha_i \rangle_{AS}$$

$$= -\frac{1}{2} y \quad (\alpha \in \text{occupied})$$

h^{HF} two body part $=$

$$\begin{bmatrix} -\frac{1}{2}y & 0 & 0 & 0 \\ 0 & -\frac{1}{2}y & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$h^{HF} = \begin{bmatrix} -\frac{1}{2}y & 0 & 0 & 0 \\ 0 & 1-\frac{1}{2}y & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$h^{HF(0)} C^{(1)} = \epsilon^{HF(1)} C^{(1)}$$

since $h^{HF(0)}$ is already diagonal:

$C^{(n)} = I = C^{(0)}$

$\epsilon^{HF(n)} = \begin{bmatrix} -\frac{1}{2}g \\ 1-\frac{1}{2}g \\ 2 \\ 3 \end{bmatrix}$

Since C is constant every iteration ϵ^{HF} and h^{HF} will be the same and we only needed 1 iteration after all.

$$E_{HF} = \sum_p \epsilon_p^{HF} - \frac{1}{2}\sum_{pq} \langle pq|\hat{V}|pq\rangle_{AS}$$

$$= 2 - 2g - \left(\frac{1}{2}\cdot 4 \cdot \left(-\frac{g}{2}\right)\right)$$

$$= 2 - g$$